

CCST9065

Our Response to Climate Change:

Hong Kong 2100

Final Portfolio Submission - Essay

"Climate Change" is a phrase that often feels strangely distant, right up until it touches your own skin. For me, that realisation hit when Hong Kong's once-reliable autumn coolness simply vanished. I have spent most of my life in densely packed concrete buildings. Now, instead of a crisp November breeze, we are left sweating through sleepless nights. The global crisis has invaded my living room.

The disappearance of our autumn is a symptom of the global problem we are facing. As temperatures rise globally, Hong Kong's endless blocks of concrete and asphalt absorb the sun, effectively turning our apartments into what can only be described as a "sauna" (Jonnyhough, 2025). As a citizen, I used to wonder if we had to tear down these megacities to fix the problem. But I realised that is not pro-development or realistic. The most sustainable approach is to work with the buildings surrounding us (Climate-KIC, 2022).

Looking at a workable solution, my key inspiration came from high school physics: thermodynamics and the reflection of heat. I believe the most practical climate action we can take in Hong Kong is urban retrofitting using high-albedo, solar-blocking coatings. Instead of feeling helpless or waiting for slow global policies to catch up, this is a physical, pro-development change we can implement right now.

High-albedo coatings are specialised reflective paints designed to maximise the reflection of solar radiation. When we use this material on the rooftops of our concrete high-rises, it intercepts the intense heat before it can soak into the building's structure. Standard concrete and dark roofing materials absorb the vast majority of the sun's energy, storing it up during the day and radiating it into our apartments at night. This trapped heat is exactly why I, like many others, find myself forced to rely on heavy air conditioning just to survive the sizzling summer nights.

However, using a solar-reflective coating can bounce up to 35% of that solar radiation away from the concrete entirely. Research analysing thermal dynamics in dense cities shows that applying these reflective surfaces can reduce indoor ambient temperatures by a highly noticeable 2 to 4 degrees Celsius (Yuan et al., 2024).

This approach perfectly echoes the circular economy thinking we explored in class. We are upgrading and extending the life of our existing infrastructure instead of demanding carbon-heavy new construction. It is also incredibly accessible. A bucket of elastomeric reflective coating is easy to find at regular hardware stores in Mong Kok or Sham Shui Po. By simply painting our roofs, we stop our residential blocks from acting like giant heat sponges, cooling our micro-communities without halting urban growth.

The biggest beneficiaries of this idea are us, specifically, local Hong Kong people. Global CO₂ levels rarely motivate people daily, but the promise of a cooler living room and a cheaper electricity bill absolutely does. Lowering our indoor temperatures by even two degrees helps a lot for the community, especially for lower-income families living in urban units who suffer the most from the heat (*Urbanization effect*. n.d.).

You do not need to wait for the government to take the first step. Today, you can draft a short proposal outlining the benefits of high-albedo roof paint and drop it into the mailbox of your building's Owners' Corporation. If you live in a village house or have a private balcony, you can go to a local hardware store this weekend, buy a tin of solar-reflective coating, and paint the sun-facing walls yourself.

We might not be able to fix global emissions overnight, but grassroots urban retrofitting proves ordinary citizens are not powerless. By reflecting the sun away from our own roofs, we can lower our bills, reclaim our comfort, and finally welcome the autumn breeze back to Hong Kong.

Citation:

- Climate KIC. (2022, December 6). *Climate KIC launches the Built Environment Circular ClimAccelerator*.
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- Jonnyhough. (2025, June 3). Thermal mass of concrete | Concrete Society. Concrete Society. <https://www.concrete.org.uk/fingertips/thermal-mass-of-concrete/>
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